

Silent Resumption: A New Test for Ellipsis

Idan Landau

Abstract

While extraction out of silent anaphors is deservedly considered one of the strongest tests for the existence of silent structure in the anaphoric site (*ellipsis*), in many environments considered to be elliptical it is not applicable, either because the ellipsis site is an inherent barrier or because it is rendered inaccessible to syntactic operations by virtue of ellipsis itself. I propose that this obstacle can be bypassed by testing whether the anaphoric site may host an internal resumptive pronoun. In a number of $\bar{A}$ -dependencies, the operator requires the presence of such a pronoun due to the *Ban on Vacuous Quantification*. This requirement holds purely at LF and is indifferent to island constraints, hence it fills the diagnostic lacuna left by the limitations of the extraction test. This new test is put to use in Hebrew, where genuine ellipsis co-exists alongside with strategies of deep anaphora like empty *n* heads and Null Complement Anaphora. The results of the silent resumption test confirm the availability of *n*P-, DP- and PP-ellipsis in the language – all categories resisting extraction out of them. Further implications are discussed that bear on the analysis of so-called “mixed anaphors” in a number of other languages.

Keywords: *ellipsis, resumption, Ban on Vacuous Quantification, mixed anaphors*

1 Introduction

It is a curious aspect of the distinction between deep and surface anaphora, introduced in the seminal work of Hankamer and Sag 1976, that although virtually everyone accepts that it captures a fundamental cut in how anaphoric expressions are represented and interpreted in natural language, much disagreement persists among scholars as to how to diagnose it in particular cases. The disagreement commonly takes two forms: An element *X* in language *L* is analyzed as a deep anaphor by some linguists or as a surface anaphor by others; alternatively, *X* is analyzed as a “mixed anaphor”, where the task is to explain why it displays the particular mixture of conflicting properties of the two types that it does. Either way, the task of classifying anaphors as deep or surface continually proves to be nontrivial.

Why is that? One reason is that some of the original criteria proposed by Hankamer & Sag (1976) do not yield the desired cut. Going over this list, Merchant (2013) maintains that in fact, only two criteria are reliable: (i) whether or not material can be extracted from the anaphoric site (overtly, producing a word-order difference, or covertly, producing inverse scope); and (ii) whether material inside the anaphoric site can trigger agreement on elements outside of it. Notably, Merchant points out that none of the other, “classic” criteria are full proof. Pragmatic control, supposedly a hallmark of deep anaphors, is available in limited cases of VP-ellipsis

(*Yes, we can! Please, do*; see Schachter 1977, 1978, Hankamer 1978, Pullum 2000, Miller and Pullum 2013). Sloppy identity occurs not only with typical ellipsis sites but also with “paycheck pronouns” and with Null Complement Anaphora (on the latter, see sections 3.1 and 4.1). And the Missing Antecedent test is unreliable insofar as it yields acceptable results not only in ellipsis cases but also with unquestionably deep anaphors (Williams 1977). Indeed, Williams went so far as to deny the reality of the deep/surface distinction (although crucially, he had no account for the “strong” criteria of extraction and agreement).

A second reason why determining status of particular anaphors as deep or surface is not obvious is the interference of independent factors or limitations of the language under consideration. Trivially, the agreement criterion is inapplicable in languages without agreement. The extraction test, in turn, is restricted in a variety of ways. Some domains, which are perfectly legitimate for ellipsis, nonetheless form absolute barriers to extraction in many languages (e.g., DP and PP; see sections 3-4). Likewise, “scope rigid” languages minimize the opportunities of detecting inverse scope readings that arise in the context of ellipsis simply because they do so *regardless* of ellipsis. Finally, derivational timing may turn ellipsis domains opaque to agreement/extraction triggered *after* the constituent has been elided, masking its surface anaphoric nature, at least with respect to some types of agreement/extraction (see Aelbrecht 2010, Baltin 2012, Johnson 2015, Sailor 2018 and the survey in Güliz and Lipták 2022).

This quick methodological assessment makes one point eminently clear: There is no surplus of reliable ellipsis criteria in the field. Quite the opposite, we still need to discover and refine additional criteria whose reliability is not easily disputed, providing us with the level of confidence accredited to the agreement/extraction tests. The goal of this paper is to introduce and explore one such novel criterion: The ability of the anaphoric site to host an internal resumptive pronoun. The testing ground will be three elliptical constructions in Hebrew, where the null constituent corresponds to an NP/*n*P, DP or PP.

As a point of departure, let us pause to ask *why* the extraction test is so reliable. Take a common case of sluicing as in (1a).

- (1) a. Ernie pounded on something, but I couldn’t tell on what.
 b. Ernie pounded on something, but I couldn’t tell
 [CP [on what]_x [TP ~~Ernie pounded x~~]].

On the standard analysis, the TP in the second clause is elided – a surface anaphor (Ross 1969, Merchant 2001). Suppose it is not; then it is a deep anaphor, lacking any internal structure. But then the stranded *wh*-phrase (*on what*), would bind no variable at LF. This runs counter to a fundamental constraint on LF representations, proposed in Chomsky 1982 (see also Koopman and Sportiche 1982 for a more restrictive version), which precludes redundant operators.

(2) *Ban on Vacuous Quantification (BVQ)*

At LF, every operator must bind some variable(s).

“Operator” in (2) refers to clause-peripheral elements in $\bar{A}$ -positions, whether quantificational (*wh*-phrases, focus phrases, raised QPs) or not (topic phrases, null operators in relative clauses, in degree clauses, in *tough* constructions etc.).

In (1b), the trace of the stranded *wh*-phrase provides the necessary variable. Without it, the BVQ would be violated. This is, then, a powerful and straightforward argument for the presence of internal structure at LF in the silent TP of sluicing. Other arguments famously obtain, like the morphosyntactic connectivity of the remnant with the trace position (e.g., the preposition *on* in the sluice of (1)), but these arguments detect *pre*-LF representations.

Factoring out the morphosyntactic evidence, which is directly linked to *movement* dependencies, we are left with the BVQ, which is not necessarily wedded to movement. In particular, it holds with equal force of *resumptive* dependencies. Before we turn to illustrate this key point, let us recall why the BVQ is an essential principle of grammar.

For Chomsky, the BVQ fell out of Full Interpretation; a variable-less operator is uninterpretable at LF. There are alternative ways of deriving it (see Potts 2002), but the choice among them is immaterial to us. Empirically, the BVQ covers a broad range of phenomena. In interrogative and in relative clauses, it forces the presence of a trace for the operator to bind (3a-b). If the dependency crosses an island, a resumptive pronoun is required (3c) – a strategy somewhat marginal in English but fully productive in other languages (the operator and the variable are set in bold below).

- (3) a. **Which armchair_i** did you like **t_i** / *the chair?
b. I actually liked **Op_i** the armchair that Mary was sitting on **t_i** / *the chair.
c. I actually liked **Op_i** the armchair that Mary was upset because **it_i**/*it_j/*the chair wasn't the right color.

Obligatory bound pronouns are found in a variety of other constructions even without islands: Left dislocation (4a), prolepsis (4b) and copy raising with a nonthematic subject (4c) (on the latter, see Landau 2009, 2011).

- (4) a. John, **Op_i** something terrible happened *(to **him_i**).
b. Mary said of her dad **Op_i** that challenges had never deterred **him_i**/*him_j/*her mom.
c. [Context: My friend describes to me how well-kept and flourishing her garden is. I remark:]
Your garden sounds like **Op_i** every day a gardener comes in *(to tend **it_i**).’

Compare (4a-c) with constructions in which the clausal constituent is not required to function as a predicate, so that no operator is generated at its edge, hence no bound pronoun is required:

Hanging topic (5a), simple clausal complementation (5b) and copy raising with a thematic subject (5c)

- (5) a. As for John, something terrible happened.
 b. Mary said to her dad that challenges had never deterred him_{i/j}/her mom.
 c. [Context: My friend takes me to see how well-kept and flourishing her garden is.
 I remark:]
 Your garden looks like every day a gardener comes in.

The BVQ has been invoked to elucidate further phenomena, like the distribution of individual-level predicates (Kratzer 1995), Sprouting (Chung et al. 1995) and the Coordinated Structure Constraint (Fox 2000). In the present study, its effects will be limited to forcing a resumptive pronoun in $\bar{A}$ -dependencies.

Resumptive pronouns are quite productive in Hebrew and have been the topic of extensive research for many decades (see, among others, Doron 1982, Borer 1984, Sells 1987, Erteschik-Shir 1992, Shlonsky 1992, Sichel 2014, Keshev and Meltzer-Asscher 2017). Two of the most common positions they are found in are possessors and prepositional objects.

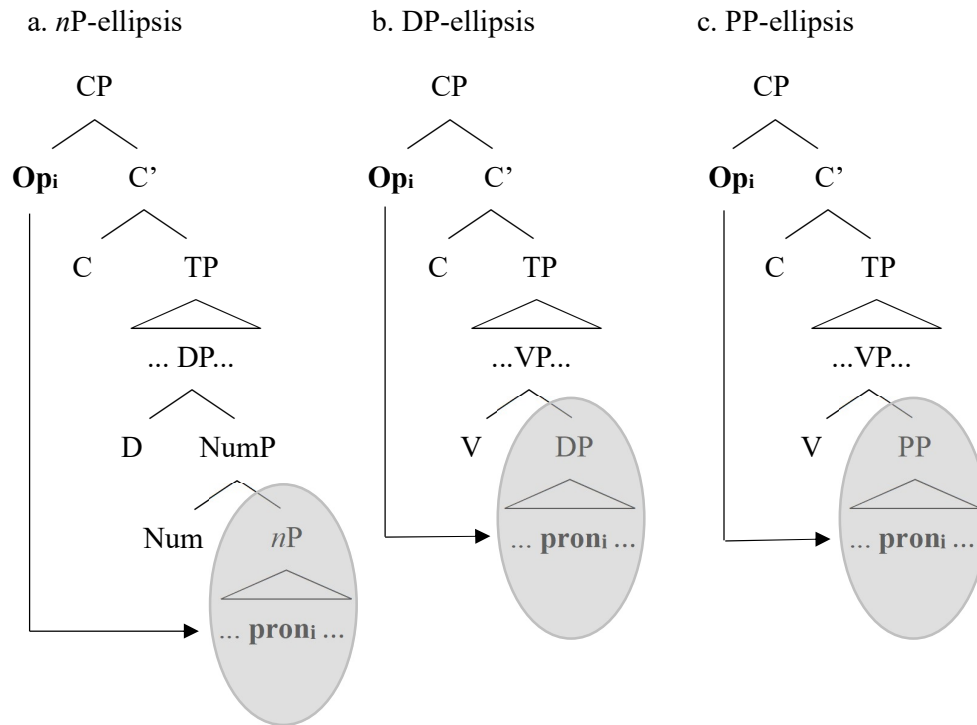
- (6) a. **Eyze bamay_i** at lo maskima lir'ot yoter sratim ***(šelo_i)?**
 which director you.F not agree.SG.F to.see more movies his
 'Which director do you no longer agree to see movies by?'
 b. yeš bamay'im **Op_i** še-ani me'od roca lesoxeax
 there.is directors that-I very.much want.SG.F to.converse
itam_i/*_i/ *im ___/*im Almodóvar.
 with.them / with / with Almodóvar
 'There are directors who I would very much like to converse with.'

In (6a), a possessive resumptive pronoun, crucially coindexed with the interrogative operator, is required inside the embedded object; clearly, the possessee noun *movies* by itself does not force the presence of the pronoun. In (6b), a resumptive pronoun is required following the embedded preposition, as Hebrew disallows P-stranding. Notice that the requirement is syntactic and not morphological; *im* 'with' is not an affixal preposition by itself (unlike *le-* 'to', *be-* 'in' and *me-* 'from'), so it needs no morphological support. Crucially, the pronoun must be coindexed with the relative operator. Thus, both cases exemplify the working of the BVQ in $\bar{A}$ -dependencies in Hebrew. Such examples will figure prominently in the following sections, *except that the resumption site will be invisible*, due to ellipsis.¹

¹ The discussion throughout assumes that resumptive pronouns, at least of the type examined here, are inserted in the syntax, hence visible to LF. It should be noted that other types of resumptive pronouns may be inserted at PF, with different consequences. Saab and Zdrojewski (2012) argue that both types of resumptive pronouns exist in River Plate Spanish: The syntactic one is found in Clitic Left Dislocation (CLLD) and the PF one in clitic doubling,

Specifically, we will examine configurations of the following abstract form.

(7) *Resumption inside ellipsis sites*



For this examination to be meaningful, the putative resumption inside the ellipsis site will be compared with attempted resumption inside *deep* anaphors; the latter is expected to fail, as deep anaphors harbor no internal structure to host the resumptive pronoun, so the operator will end up violating the BVQ. Taken together, such contrastive pairs constitute what I call the Ellipsis-Internal Resumption test, or in short, EIR.

Two sets of considerations underscore the theoretical significance of this novel test for ellipsis. As noted above, there is general consensus that ellipsis is best demonstrated by extraction. However, in quite a few ellipsis configurations, extraction is unavailable for a variety of independent reasons. This state of affairs has forced much of the literature on NP-ellipsis and Argument Ellipsis to focus on *interpretive* properties of the null position rather than on its *syntactic* properties; yet such arguments are inherently weak for the reasons already mentioned above (see section 2.1 for detailed discussion).

which optionally occurs with focus fronting. Interestingly, under ellipsis, the former one continues to repair (certain) islands, but the latter fails to do so. Saab and Zdrojewski reason that ellipsis (applying at PF) bleeds the insertion of the PF clitic, leading to the anti-repair effect. The behavior of the Hebrew resumptive pronouns clearly indicates that they are visible at LF, hence cannot be inserted at PF.

A second, unfortunate circumstance is that although some categories are not inherent barriers for extraction, in point of fact their transparency to extraction is greatly diminished by ellipsis. These are the so-called “mixed anaphors”, which display a bewildering selective transparency to different kinds of extractions: Danish *det* (Hauser et al. 2007), Dutch Modal Complement Anaphora (Aelbrecht 2010), British English *do* (Baltin 2012), Norwegian *gjøre det* (Bentzen et al. 2013), Swedish *det* (Thompson 2014), English *do so* (Bouton 1969, Houser 2010, Kim 2016, Park 2018, Breuning 2019) and Dutch *dat doen* (Heynen 2022).

Although the crosslinguistic picture is not uniform, an overarching tendency is for $\bar{A}$ -extraction to be blocked from inside the very same anaphors that allow A-extraction. There are a number of theoretical responses to this recurring pattern; some of them, in fact, deny any internal structure to the anaphor, i.e., take it to be a deep anaphor. Enter the EIR test. Resumptive dependencies are a type of $\bar{A}$ -dependency. If some of the anaphors that resist $\bar{A}$ -extraction nonetheless *allow* the formation of a resumptive dependency terminating inside the anaphor, this will furnish a strong argument against the deep anaphor analysis (while leaving the ban on extraction in need of some independent explanation). At the same time, such a contrast would lend support to a timing-based approach to the ban on $\bar{A}$ -extraction from the ellipsis site, insofar as the critical difference between movement and resumption concerns the derivational stage at which the dependencies are registered. Section 5 takes a glimpse at these facts and their implications.

The interaction of ellipsis (most often, sluicing) and resumption has not gone unnoticed in the literature, but it is usually discussed in the context of its implications for island repair; that is, a context *presupposing* rather than arguing for surface anaphora. Thus, Wang (2007) and Boeckx (2008) attributed the apparent island-insensitivity of sluices with indefinite correlates to the presence of a resumptive pronoun inside the sluice. While there are compelling reasons to reject this analysis for English (Merchant 2001, Rottman and Yoshida 2013, Barros et al. 2014), it may be true for contrast sluices in other languages, where resumption enjoys wider distribution (Abels and Thoms 2014). Nonetheless, the importance of resumption in *establishing* ellipsis has been observed in Saab 2010. Saab argued that TP-ellipsis in Spanish, where the remnant is accompanied by a polarity particle, can host the internal resumptive clitic of Clitic Left Dislocation (CLLD) (8a), and in that respect it differs from *hacerlo* ‘do it’ anaphora (8b), which is a true deep anaphor with no internal structure.²

- (8) a. Juan desaprobó a María, pero a Ana_i, no ~~la_i~~——desaprobó.
 Juan failed DOM María but DOM Ana not CL-ACC-3SG.F failed
 ‘Juan failed MARÍA, but not Ana.’

² A reviewer observes that (8a-b) do not form a minimal pair, as the antecedent in (8b) contains both a fronted object and a clitic, whereas the one in (8a) contains neither. The point is well-taken, but probably does not detract from the import of Saab’s argument, since the syntactic parallelism is even tighter in (8b) than it is in (8a), and yet it does not produce grammaticality (because *hacerlo* ‘do it’ cannot host the bound clitic).

- b. *A María la besó Juan y, a Ana, lo hizo Pedro
 DOM María, CL-ACC-3SG.F kissed Juan and DOM Ana CL.N did Pedro
 (Intended: ‘María, Juan kissed and Ana, Pedro did.’)

The contrast in (8) is formally equivalent to the contrasts to be studied below in Hebrew. While the status of TP-ellipsis as a surface anaphora in Spanish is well-established, the status of the null categories NP/DP/PP categories in Hebrew is far from obvious and demands close scrutiny. Furthermore, whether CLLD is derived by movement or not (Saab, in fact, assumed the former) is a contested issue in the literature. The resumption constructions examined here, however, are most likely not derived by movement. This point is indeed contested in the literature on resumption on Hebrew, so I address it separately in the Appendix, where it is shown that reconstruction data do not establish a movement-derivation for Hebrew resumptive pronouns.

The paper is organized as follows. Section 2 discusses null *n*Ps (traditionally called null NPs). Section 2.1 presents the challenges of distinguishing deep and surface anaphora in this domain; chief among them are the unreliability of the strict/sloppy reading test and the difficulties in extraction out of *n*P. Section 2.2 applies the EIR test, establishing that Hebrew indeed can generate null *n*P either as deep or as surface anaphors. Section 3 discusses null DPs, more commonly known as Argument Ellipsis (AE). Section 3.1 presents the challenges of distinguishing deep and surface anaphora in this domain; specifically, how to rule out a *pro* or NCA analysis of a missing argument, again under the limitations imposed by the strict/sloppy reading and the extraction tests. Section 3.2 demonstrates resumption inside null DPs using the EIR test, providing a novel argument for AE in Hebrew. Section 4 turns to the little-discussed case of null PPs. Section 4.1 presents the challenges of establishing that PPs ever undergo ellipsis, given the ubiquitous option of NCA (implicit PPs). Section 4.2 shows that even null PPs, in contexts supporting ellipsis, allow internal resumption. This is a novel argument for AE of PPs – not only in Hebrew but in general, too. Section 5 examines some of the implications of ellipsis-internal resumption for some contested cases of deep vs. surface anaphora in the broader literature outside Hebrew. Section 6 concludes the paper.

2. NP-ellipsis

In section 2.1 I present the current common understanding of the two ways in which nominal nuclei can be missing: a null *n* head (deep anaphora) and an elided *n*P (surface anaphora). We will see that in Hebrew, the two strategies cannot be distinguished by the inherent semantics of the null nominal. Nor can they be distinguished by extraction, which is barred out of DPs in the language. The single reliable distinction is in the projection of argument structure – licensed by the root inside *n*P but not by a bare *n*. Section 2.2 shows how ellipsis-internal resumption reliably strengthens the distinction. Nouns that can go missing as deep anaphors (that is,

identified by a *deictic* antecedent) resist omission when placed in construction with an operator, which needs to bind a variable. However, given a *linguistic* antecedent, they can emerge as surface anaphors, and then the operator is licensed, as depicted in (7a).

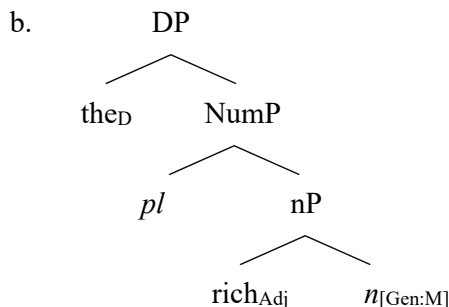
2.1 The challenge of establishing NP *ellipsis*

DPs often surface with their nominal nucleus missing, sometimes along with its arguments and some adjuncts, leaving only peripheral overt material (demonstratives, possessors, modifiers, numerals, quantifiers, etc.). A long-standing observation in the study nominal syntax is that languages regularly employ at least two methods of arriving at these “incomplete” surface constructions, that is, two kinds of null NPs, roughly corresponding to the classical distinction between deep and surface anaphors (Hankamer and Sag 1976). The first kind, which I will call *empty nouns* (EN), corresponds to the deep anaphor: It can be used without a linguistic antecedent, where the content of the EN is pragmatically recovered; it is often semantically restricted, e.g., to humans, to abstract entities or to generic contexts; and it has no internal syntactic structure. Conversely, the second kind, which I call *elided noun phrase* (ENP), corresponds to the surface anaphor: It can only be used with, and be interpreted as, a salient linguistic antecedent; it is semantically unrestricted; and it has the normal internal syntactic structure displayed by NPs (see Saab 2019 for a useful recent comparison between the two kinds of null NPs).

What are the structures of null NPs generated as deep anaphors and of those generated as surface anaphors? I follow recent literature in assuming that *all* null NPs are minimally *n*Ps, that is, functional projections encoding the nominal category, gender, and possibly other inherent features (Corver and van Koppen 2011, Merchant 2014, 2023, Saab and Lipták 2016, Saab 2019). In the case of deep anaphors, a null bare *n* head, unaccompanied by a root or internal arguments, is merged as the complement of D. The denotation of this null *n* is a general concept drawn from a universal, very restricted set; e.g., PERSON, THING, TIME, PLACE, etc. Common practice also assumes that *n* is the locus of gender features (Kramer 2016).

(9) *Empty Noun (EN): Null n as a deep anaphor*

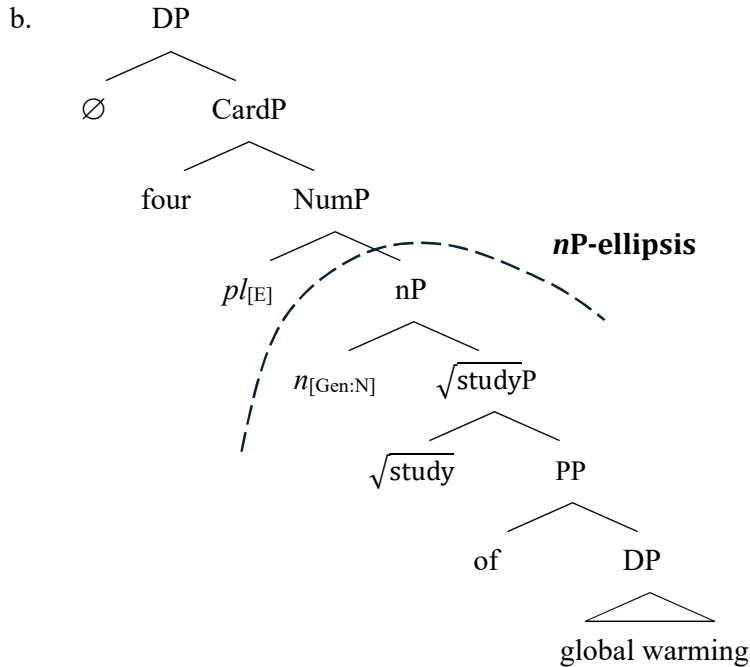
a. The rich do not pay taxes like you and me.



In the case of surface anaphors, *nP* undergoes ellipsis under identity with a linguistic antecedent. Crucially, a full nominal structure – root plus argument structure – is present inside the elided *nP*. We may assume that the [E]-feature licensing ellipsis is hosted on Num.

(10) *NPE (NP ellipsis): Null nP as a surface anaphor*

a. Bill read three studies of global warming and Matt read four.



Prima facie, then, it seems that ENs and ENPs are sufficiently distinct to be easily distinguishable in any language. The fact of the matter, though, is that the distinction is not so easy to draw, primarily because in most contexts, DPs lacking their nominal nucleus lend themselves to either analysis. Furthermore, occasionally some of the crucial empirical tests are unavailable for independent reasons. This state of affairs leaves the distinction standing on a rather tenuous factual basis as far as particular languages are concerned.

Consider first semantic restrictions. In some languages, ENs in many contexts must refer to humans. This is often the case in English, which further restricts human ENs to generic contexts (Kester 1996, Panagiotidis 2003).

- (11) a. Everybody admires the tall.
 [= *Everybody admires tall people*
 ≠ *Everybody admires tall things/buildings*]
- b. [Pointing to two guys:] I know the tall *(one).
- c. [Pointing to two floor lamps:] I prefer the tall *(one).

However, Hebrew ENs seem to be free of these restrictions; the versions without the parenthesized nouns below represent the EN options.³

- (12) a. kulam ma'aricim (anašim) gvohim.
 Everybody admire.PL.M people(PL.M) tall.PL.M
 'Everybody admires tall people / the tall.'
- b. [Pointing to two guys:]
 ani makir et (ha-baxur) ha-gavoha.
 I know.SG.M ACC the-guy(SG.M) the-tall.SG.M
 'I know the tall guy / one.'
- c. [Pointing to two floor lamps:]
 ani ma'dif et (ha-menora) ha-gvoha.
 I know.SG.M ACC the-lamp(SG.F) the-tall.SG.F
 'I prefer the tall lamp / one.'

In general, there does not seem to be a lexical noun in Hebrew that cannot be omitted in contexts supplying enough deictic information to recover its content.

A second test to distinguish EN from NPE does yield useful results in Hebrew. Recall that ENs are the silent counterparts of concepts drawn from a very restricted set – PERSON, THING, TIME, PLACE, and maybe a few more. These nouns do not select or project arguments. In particular, although an event (e.g., *performance*, *cooking*, *plan*) can fall under the very broad description THING, for any of the participants in the event to appear as syntactic constituents, its nominal nucleus cannot be the EN THING itself, but rather a particular lexical nominalization endowed with argument structure. In this light, compare NPE in (13a) and EN in (13b). The context for both is identical: We sit in a music class, listening to and comparing different, multiple renditions of two sonatas by Beethoven, no. 11 and no. 12.

- (13) a. NPE
 A: ha-bicua ha-rišon šel mispar 11 haya haxi mavrik.
 the-performance the-first of number 11 was most brilliant
 'The first performance of No. 11 was the most brilliant one.'
- B: aval ze/ha-šeni šel mispar 12 haya haxi merageš.
 but that/the-second of number 12 was most moving
 Lit. 'But that/the second ____ of No. 12 was the most moving one.'

³ The [human] restriction does characterize a small set of nouns, homophonous with adjectives, like *xaxam* 'sage', *xole* 'patient', *karov* 'a kin, a relative'. As Borer and Roy (2010) show, these nouns display idiosyncratic meanings that are not predictable from the related adjectives. As they are nouns themselves, no EN occurs with them, so I set them aside.

b. *EN*

[After the playing of the last performance ends:]

ze/ha-šeni (*šel mispar 12) haya haxi merageš.

that/the-second of number 12 was most moving

Lit. ‘That/the second (*of No. 12) was the most moving one.’

In (13a), the subject DP in B’s utterance is elliptical; the nominal nucleus *bicua* ‘performance’ is missing, but everything else remains: a demonstrative pronoun or an ordinal numeral, and the internal argument. How can the internal argument survive ellipsis of *nP*? Presumably, it is introduced as a complement of the root, inside that *nP*. Two options have been explored in the literature. The first one, by Yoshida et al.’s (2012), who analysed parallel constructions in English (*John read Bill’s book of poems and Mary’s ___ of music*), is a “pseudogapping” derivation: The internal argument evacuates *nP* by rightward adjunction to an intermediate functional projection within the DP, thus escaping deletion (see Saab 2019 for the same analysis applied to Spanish). A second option is to make focus-marked elements within an ellipsis domain exempt from deletion (Benz et al. 2024). For present purposes, the choice between them is not important.

In sharp contrast, (13b) is uttered without any linguistic context. Therefore, the missing nominal nucleus cannot be recovered as the descriptive noun *bicua* ‘performance’, and can only be construed as the most appropriate EN, namely THING. Given that ENs do not introduce arguments, the internal argument is not licensed.

If arguments are present in NPE but not in EN, we expect them to surface not only inside the target nominal but also outside of it, if extracted. ENs, however, having no internal structure, should block extraction. This leads to our third test. Indeed, the English *one*-anaphor, which corresponds to a deep anaphor, blocks extraction (14), but NPE allows it (15) (Merchant 2023).⁴

(14) a. * This is the battle on which the commission produced two long reports, and this is the one on which they produced three short **ones**.

b. * Taxes are something that it’s easy to find uninformed objectors to – it’s infrastructure that’s hard to find many informed **ones**.

(15) a. This is the battle on which the commission produced two long reports, and this is the one on which they produced three ____.

⁴ A reviewer disagrees with Merchant’s judgments, finding (14a) grammatical and (15b) ungrammatical. The licensing of ellipsis famously exhibits arbitrary distinctions, and speakers may well differ in which functional heads they endow with the [E]-feature. The reviewer’s idiolect can be described as allowing NPE under *one* in principle, but [E] only occurs on Card⁰ (like *three*) and not on Q (like *many*), thus licensing NPE in (14a)/(15a) but not in (14b)/(15b).

- b. Taxes are something that it's easy to find objectors to – it's infrastructure that's hard to find many ____.

Merchant (2023) and Saab (2019) present similar extraction data from NPE in German, Greek and Spanish. In languages like Hebrew, however, the extraction test is hardly usable, as DPs of any kind – definite or indefinite, count or mass – are islands (note that P-stranding is absolutely barred in Hebrew).⁵

- (16) a. * [šel mi]_i hu ciyer [šloša dyokna'ot t_i]?
 of who he painted.3SG.M three portraits
 'Who did he paint three portraits of?'
 b. * [la-pitaron ha-ze]_i Dan makir [harbe minagdim t_i].
 to.the-solution the-this Dan knows.SG.M many objectors
 'This solution, Dan knows many objectors to.'

The methodological point is broader and holds across many languages, as Saab 2019:535 recognizes: “In the nominal domain, however, there is no easy way to construct this type of example because extraction out of DPs requires manipulating too many variables (e.g., the relative position of the DP, its definiteness/specificity, among other poorly understood variables.”

In sum: While the EN-ENP distinction is well established at the theoretical level, identifying reliable empirical correlates of this distinction is not equally easy in different languages. As we have seen, in Hebrew only the argument-taking property yields useful results. While many cases where a nominal nucleus is missing under identity with an antecedent could be analysed as ENP, nothing rules out an EN analysis of these very same cases. Therefore, adding a novel syntactic diagnostic that can tease apart the two analyses would be welcome.

2.2 Resumptive pronouns inside NPE sites

As noted in the previous section, nominal expressions in spoken Hebrew are commonly used without their nominal nucleus, surfacing as a bare quantifier, numeral or attributive adjective. The content of the “missing” noun is then recovered from the conversational context. Crucially, an explicit linguistic antecedent is not required, indicating that at least these cases involve ENs.

⁵ Landau (2018:ex. 50) cites a rare exception, involving Argument Ellipsis of an in-situ unaccusative subject, from which subextraction has applied. Differently from DPs, CP complements regularly allow extraction, and elided CPs are no different (Landau 2023).

- (17) a. [Context: John shows me the first album of “Queens of the Stone Age”, and then shows me the second one. I say:]
 ahavti et ha-šeni.
 liked.1SG ACC the-second
 ‘I liked the second one.’
- b. [Context: Looking at a pile of Continental Philosophy books on my friend’s desk, I sigh and say:]
 maspik li likro rak xelek.
 enough to.me to.read only part
 ‘It’s enough for me to read only part (some) of these.’
- c. [Context: In front of the wine bar, I’m having a hard time selecting my wine, so the bartender proposes:]
 ha-lavan mumlac.
 the-white recommended
 ‘The white one is recommended.’
- d. [Context: Opening my mailbox after a long time, I’m surprised to find so many letters, and react:]
 Huh, kiviti le-šloša.
 huh hoped.1SG to-three
 ‘Huh, I was (only) hoping for three of them.’

As one can see, ENs are licensed inside different kinds of DPs in Hebrew. The remnant may be an ordinal numeral (17a), a quantifier (17b), an attributive adjective (17c) or a cardinal numeral (17d). In fact, any free-standing dependent of N can occur without an overt N, in the right context.⁶

⁶ Phonologically dependent elements, like the head of a construct state, may not occur without an overt noun. Thus, the cardinal *two*, which must appear as a construct head (*šney/štey*, for MASC/FEM) in combination with an overt noun (i), shifts to a free-standing form (*šnayim/štayim*) when occurring with a null N (ii). As Doron (2013: ex. 9) observes, the construct form of the numeral is licensed by other overt remnants, such as attributive adjectives. Interestingly, in this case it alternates with the free-standing form (iii).

- (i) hu hevi šaloš mizvadot, ani heveti **štey/*štayim** mizvadot.
 he brought.3SG.M three suitcases I brought.1SG two suitcases
 ‘He brought three suitcases, I brought two suitcases.’
- (ii) hu hevi šaloš mizvadot, ani heveti **štayim/*štey**.
 he brought.3SG.M three suitcases I brought.1SG two
 ‘He brought three suitcases, I brought two.’
- (iii) hu hevi šaloš mizvadot kalot, ani heveti **štayim/štey** kvedot.
 he brought.3SG.M three suitcases light I brought.1SG two heavy
 ‘He brought three light suitcases, I brought two heavy ones.’

Inflectional “strengthening” of determiners and adjectives modifying a null N is a familiar pattern crosslinguistically (Corver and van Koppen 2011, Saab and Lipták 2016). Within a grammatical architecture assuming Late Insertion, the phonological shape of terminal nodes is determined at PF, and may well occur *after*

Things are dramatically different if the EN hosts a resumptive pronoun, which functions as the sole variable for an $\bar{A}$ -operator to bind. Because the BVQ constrains *syntactic* representations, this resumptive pronoun must be syntactically present. The following sentences embed (17a-d) inside a restrictive relative clause (18a,b), an interrogative sentence (19c) and a free relative clause (20d). The results are ungrammatical in the above context or in any other imaginable context. This is readily understood if ENs, which are perfectly grammatical in non- $\bar{A}$ -binding environments like (20a-d), are simplex deep anaphors (specifically, *n* heads) with no internal structure to host the resumptive pronoun.

- (20) a. *yeš la'hakot še-ahavti et ha-šeni.
 there.exist bands that-liked.1SG ACC the-second
 ('There are bands that I liked the second one.')
- b. *yeš mexabrim še-maspik li likro rak xelek.
 there.exist authors that-enough to.me to.read only part
 ('There are authors that it's enough for me to read only part of these.')
- c. *eyze yekev ha-lavan mumlac?
 which winery the-white recommended
 ('Which winery is the white one recommended?')
- d. *mi še-kiviti le-šloša ba-sof katvu harbe yoter.
 who hoped.1SG to-three in.the-end wrote.3PL a.lot more
 ('Who I was hoping for three of these eventually wrote a lot more.')

We now come to the crucial prediction. If null NPs can be generated *either* as unstructured deep anaphors *or* as surface anaphors, namely, a full NP structure that undergoes ellipsis, then the latter structure *should* be able to host an internal resumptive pronoun. Correspondingly, contexts that allow ENP, by providing a local NP antecedent, should not give rise to BVQ violations, in contrast to (20a-d). The prediction is fully confirmed. The same sentences are grammatical given the appropriate linguistic context; furthermore, the operator in the elliptical clause is indeed interpreted as binding a variable inside the ellipsis site.

ellipsis (which is licensed already in the syntax). Thus, it is not surprising to find the latter feeding and affecting the former.

- (21) a. *yeš la'hakot še-ahavti et **ha-albom** ha-rišon **šela'hem**,*
 there.exist bands that-liked.1SG ACC the-album the-first their
 and-ve-yeš la'hakot še-ahavti et ha-šeni.
 there.exist bands that-liked.1SG ACC the-second
 ‘There are bands that I liked their first album, and there are bands that I liked their
 second one.’
- b. *yeš mexabrim še-xašuv li likro et kol **ha-sfarim** **šela'hem**,*
 there.exist authors that-important to.me to.read ACC all the-books their
 ve-yeš mexabrim še-maspik li likro rak xelek.
 and-there.exist authors that-enough to.me to.read only part
 ‘There are authors that it’s important for me to read all their books, and there are
 authors that it’s enough for me to read only part (some) of their books.’
- c. *eyze yekev **ha-yain** ha-adom **šela'hem** mumlac,*
 which winery the-wine the-red their recommended
 ve-eyze yekev ha-lavan mumlac?
 and-which winery the-white recommended
 Lit. ‘Which winery is their red wine recommended, and which winery is
 their white one recommended?’
- d. *mi še-kiviti le-xamiša **mixtavim** **mimeno**,*
 who that-hoped.1SG to-five letters from.him
 ve-mi še-kiviti le-šloša, ba-sof katvu harbe yoter.
 and-who that-hoped.1SG to-three in.the-end wrote.3PL a.lot more
 Lit. ‘Who I was hoping for five letters from him and who I was hoping for three
 letters from him eventually wrote a lot more.’

The diversity of these environments is noteworthy. The semantic function of the operator (restrictive, maximizing, interrogative) is not relevant, as long as it needs a variable to bind. The position of the variable is also immaterial, as long as it is syntactically present. Notice that resumption most commonly occurs with possessive pronouns, but this is not necessary; oblique PP may also host a resumptive pronoun, showing the same behavior (21d).

Taking representations (9)-(10) as our model, we can provide an explicit syntactic explanation for the contrast. The null nominals in (17)-(20), which have no linguistic antecedent, are generated as ENs, on a par with (9). Those in (21), which *do* have linguistic antecedents, are generated as NPEs, on a par with (10). The presence vs. absence of internal structure inside the null NP then determines whether the formation of a variable binding relation with resumptive pronoun inside the null NP is possible or not; when it is not, the structure is ruled out by the BVQ. To illustrate, the very same sentence is parsed with EN in (20b) and with NPE in (21b), as shown in (22a,b), respectively (English words used for convenience).

popularity”, Tomioka (2014:253) remarks of this test, “it is perhaps the most problematic.” Let us see why.

This test is indeed straightforward in the most elementary cases. In (23a), the pronoun *oto* in the second conjunct very strongly favors the strict reading, while the object gap in (23b) readily allows the sloppy reading as well. This is taken as evidence for the presence of a surface anaphor in the latter, namely, the result of AE.

- (23) a. Ayelet_i ohevet et ha-bayt šelai_i, aval Gali lo ohevet oto.
 Ayelet likes.SG.F ACC the-house her but Gali not likes.SG.F it
 ‘Ayelet_i likes her_i house but Gali doesn’t like it.’
 [*it* = ✓Ayelet’s house, ?? Gali’s house]
- b. Ayelet_i ohevet et ha-bayt šelai_i, aval Gali lo ohevet Ø.
 Ayelet likes.SG.F ACC the-house her but Gali not likes.SG.F
 ‘Ayelet_i likes her_i house but Gali doesn’t.’
 [Ø = ~~her house~~ = ✓Ayelet’s house, ✓Gali’s house]

However, slight modifications in the antecedent easily lifts the sloppy reading of an overt pronoun to a perfectly acceptable level. These modifications include (i) tight pragmatic association between the local subject and the possessive pronoun (24a); (ii) a conjoined antecedent for the antecedent possessive pronoun (24b); (iii) a quantificational antecedent for the antecedent possessive pronoun (24c). Notice that the choice of the singular pronoun *ota* ‘it.F’ in (24b) *forces* the sloppy reading, since it cannot refer to the antecedent *plural* DP. Likewise, *oto* ‘it.M’ (24c) must receive a sloppy interpretation as there is no unique referent for the antecedent, which contains a bound variable, to support a strict reading.

- (24) a. Ayelet_i ohevet et ha-tafkid šelai_i, aval Gali lo ohevet oto.
 Ayelet likes.SG.F ACC the-role her but Gali not likes.SG.F it
 ‘Ayelet_i likes her_i role but Gali doesn’t like it.’
 [*it* = ✓Ayelet’s role, ✓Gali’s role]
- b. [Ayelet, Tomer ve-Roni]_i hidpisu et ha-matalot šelahem_i,
 Ayelet Tomer and-Roni printed.3PL ACC the-assignments their
 aval Yahli lo hidpis ota/otan.
 but Yahli not printed.3SG.M it/them
 ‘Ayelet, Tomer and Roni printed out their assignments but Yahli didn’t print it out.’
 [*it* = * Ayelet, Tomer and Roni’s assignments, ✓Yahli’s assignment;
them = ✓Ayelet, Tomer and Roni’s assignments, ✓Yahli’s assignments]

- c. A: lama af exad lo nika et ha-xeder šelo?
 why no one not cleaned.3SG.MACC the-room his
 ‘Why didn’t anyone clean their rooms?’
 B: ani nikiti oto!
 I cleaned.1SG it
 ‘I did!’

Pronouns permitting sloppy readings are, of course, nothing new in the semantic literature, and have been studied ever since Geach 1962 under the titles of “pronouns of laziness”, “paycheck pronouns” and “E-type pronouns” (Karttunen 1969, Partee 1972, Evans 1977, Cooper 1979, Heim 1990, Jacobson 2000, Elbourne 2001, 2008). Broadly speaking, there are two approaches to their analysis. The semantic approach assigns them a complex semantic type, a function mapping the local antecedent (e.g., *Yahli* in (24b)) to the associated referent (*Yahli’s assignments*). The syntactic approach takes the surface pronoun to be a proxy for a full syntactic NP structure, which undergoes ellipsis, and it is the internal pronoun inside this silent NP that accounts for strict/sloppy ambiguity. Crucially, both approaches rely on pragmatic salience for the licensing of the referential “middleman” (the semantic function or the elided NP); thus, both recognize that some contexts (like (24)) are better at facilitating the E-type interpretation than others (like (23)).

The implications of these observations for studies of AE are significant. The very existence of a sloppy interpretation of a null argument is no reliable proof for an underlying *surface* anaphora. In truth, such readings appear with overt pronouns, so there is no a priori basis for excluding them from null pronouns. Indeed, this point is explicitly defended in Tomioka 2003, 2014, and similar considerations led Merchant (2013:540) to conclude: “The presence of sloppy identity readings is not a diagnostic for ellipsis.”

Specifically, if E-type readings indeed emerge from an underlying elided NP, as the syntactic approach holds, then nothing is at fault with the logic of the AE studies using the sloppy reading test. If, however, the semantic approach to E-type pronouns is the right one, then the sloppy readings attested with null objects may well arise from the same semantic mechanism applying to *pro*. In that case, the inference from sloppy readings to AE is invalid. Since the literature on E-type pronouns has not delivered a clear verdict on which approach is the right one, we are left with a test whose usefulness for establishing AE (or DP ellipsis) is dubious.

In fact, at a purely empirical level, the strict/sloppy test has proven somewhat unsystematic (as the fluid distinction between (23a) and (24a) already suggests). Revisiting earlier studies of AE in Turkish and Chinese, Simpson et al. (2013) pointed out that null subjects in these languages

can receive sloppy interpretations, although variation among speakers is significant. They also observed that null subjects in Hindi and Bangla, while normally rejecting the sloppy reading, nonetheless allow it given a rich pragmatic context. Notice that these findings pertain to the interpretation of *pro*, but they bear the hallmark of E-type pronouns – sensitivity to context.

The final blow to the argument for AE from sloppy readings consists in an observation made by Cheng (2013:127): Null arguments that are deep anaphors also allow it.

- (25) [Context: Zhangsan and Lisi each owns a car. Zhangsan is washing his own car outside and Lisi sees this. Lisi says:]
 Wo zuotian xi-guo-le ____ *Chinese*
 I yesterday wash-PERF-INC.
 ‘I’ve washed my/your car yesterday.’

Hebrew is no different. The following example pragmatically favors the sloppy reading; it also illustrates that “sloppiness” is not tied to 1st/2nd person pronouns.

- (26) [Context: In Bill’s family, everyone cleans their rooms. Bill’s mom is with a friend in the living room. They see Bill coming out of his room with a broom and a rag. Bill’s mom says to her friend:]
 hu ha-axaron. kulam kvar niku ____.
 he the-last everyone already cleaned.3PL
 ‘He’s the last one. Everyone has already cleaned their rooms.’

Cheng (2013) maintained that the sloppy reading in (25) is derived from a null topic strategy: the topic *self’s car* undergoes $\bar{A}$ -movement and subsequent deletion, and the null object is its trace. This account is not applicable to Hebrew, where compelling arguments have been given against deriving null objects in this manner (Erteschik-Shir et al. 2013, Landau 2018). It seems that NCA is capable of generating these readings no less than pronouns do.

These observations dovetail with much older observations, which indicate that deep VP anaphors (*do it, do the same*) also support sloppy readings (Hobbs and Kehler 1997).

- (27) a. John revised his paper before Bill did it.
 b. John revised his paper and Bill did the same.

Of course, the Chinese/Hebrew facts are even more compelling insofar as an anaphoric element with the *very same shape* – namely, $[e]_{DP}$ – is used either as a deep anaphor or as a surface anaphor, and both guises demonstrably support sloppy readings.

To sum up: The greater tolerance of null objects to sloppy readings compared to overt pronouns is gradient and fluid. While it could reflect an underlying syntactic difference – surface vs. deep anaphora – this inference is not compelling, especially in light of non-elliptical null objects that are perfectly compatible with sloppy readings.

Are there are other interpretive differences between null objects and overt pronouns that might favor the AE analysis? The literature on AE commonly indicates that null objects can receive quantificational readings, which are, presumably, not expressible by pronouns.⁹ However, this claim has been met with skepticism. Tomioka (2014) observes that *weak* quantificational readings do not require a genuine $\langle\langle e, t \rangle, t \rangle$ denotation and can be modeled by property anaphora (= *pro* _{$\langle e, t \rangle$}) and existential closure. Kurafuji (2019) and Landau 2025 to appear propose instead to model such readings with choice functions. Landau goes on to show that truly quantificational readings are actually *unavailable* in AE sites; whenever a paraphrase with an E-type pronoun or a choice function is impossible (e.g., with narrow-scope universals or with downward-entailing quantifiers), AE is blocked. Once again, interpretive evidence for AE is rather weak, or at least equivocal, upon closer inspection.

Finally, consider nonspecific and disjunctive readings, shown respectively in (28a-b) (from Landau 2018). Such readings reject paraphrases with definite pronouns.¹⁰

- (28) A: lo niš'ar li kesef.
 not remained to.me money
 'I have no money left.'
 B: li niš'ar ____ /*oto.
 to.me remained *it
 'I have some money left.'

- (29) A: cilamti knesiya o katedrala, ani lo batuax.
 photographed.1SG church or cathedral I not sure
 'I photographed a church or a cathedral, I'm not sure.'

⁹ See, among others, Takahashi 2008, Şener and Takahashi 2010, Cheng 2013, Simpson et al. 2013, Sato 2014, 2015, Sato and Karimi 2016, Landau 2018, Maeda 2019, Han et al. 2020, Sakamoto 2020.

¹⁰ On indefinite/nonspecific AE, see Giannakidou and Merchant 1997, Hoji 1998, Kurafuji 2019, Soltan 2020. On disjunctive AE, see Funakoshi 2013, Sakamoto 2015, 2016, Lee 2016, Cyrino and Lopes 2016.

B: gam ani cilamti _____. *Disjunctive reading*
 also I photographed.1SG
 ‘I also photographed a church or a cathedral.’

B': gam ani cilamti ota. *E-type reading*
 also I photographed.1SG it
 ‘I also photographed the one that you did.’

Do these readings prove that ellipsis had applied? It is, in fact, possible that indefinite pronouns such as the English *one* have null counterparts in other languages and these null indefinite pronouns account for cases like (28B). Once again, this will be an ellipsis-free explanation, making use of deep anaphora only. To illustrate, consider (30).

(30) [Context: I see my friend fumbling for a cigarette in his pocket but finds none. I say:]
 niš'aru ____ li ba-tik, ani avi ____ lexa.
 remained.3PL to.me in.the-bag I will.get.1SG to.you
 ‘I have some left in my bag, I’ll get you.’

This is a case of deep anaphora; the understood indefinite noun *sigaryot* ‘cigarettes’ is not present in the linguistic context. In fact, it is not even present in the *actual* deictic context; its salience is purely conceptual, based on the speaker’s inference as to what his friend is looking for. If languages harbor devices for making such referential acts with no recourse to ellipsis, then the mere existence of nonspecific readings for null objects that *do* have linguistic antecedents is no compelling argument for AE.

What about the disjunctive reading of the null object in (29)? As far as I know, there is yet no alternative to AE that yields this reading, so we may take it as solid evidence, for now (although see Ahn & Cho 2020 for the claim that pronouns in Korean can support disjunctive readings). Overall, though, the incontrovertible evidence for AE coming from semantics is significantly narrower than what is commonly declared.

For this reason, the field is in dire need of reliable *syntactic* evidence. Once again, this is where the extraction test would be distinctly valuable. However, two factors conspire to make the test unavailable in many of the relevant languages. The East Asian languages severely restrict overt extraction out of AE sites (Sakamoto 2020), while in languages like Hebrew, DPs are nearly absolute islands for extraction, regardless of ellipsis. Of course, many of the languages in which extraction from DP is attested and extensively studied (e.g., Italian, Spanish, German) simply lack AE.

A rare illustration that extraction is possible out of elided DPs comes from Brazilian Portuguese (Panitz 2023:36).

- (31) A: Foi [do Pelé]_i que a Maria vendeu [DP a foto t_i] pra mim.
 was of.the Pelé that the Maria sold the picture to me
 Lit. ‘It was of Pelé that Maria sold the picture to me.’
 B: Não, foi do Kaká que ela vendeu ___ pra você.
 no, was of.the Kaká that she sold ___ to you
 Lit. ‘No, it was of Kaká that she sold the picture to you.’

One would like to see more such data from languages with AE. In the meantime, we can turn our attention to the possibility of resumption inside null objects, which is plausibly more widespread, hence can usefully establish DP ellipsis in more languages where the distinction between deep and surface anaphora in this domain is often subtle or blurred. I return in section 5 to discuss how data of this type can fruitfully inform the debate within Korean linguistics as to the proper analysis of null objects in that language.

3.2 Resumptive pronouns inside DPE sites

The following examples demonstrate that direct objects in Hebrew can be realized as null deep anaphors whose content is retrieved from the deictic context. Such deep anaphors can be analyzed in a variety of ways – as Null Complement Anaphora (NCA), *pro*, or $\bar{A}$ -variable. I will assume the NCA analysis here without attempting to defend it; the choice among these options is actually not crucial insofar as they are all *non*-surface anaphors.

- (32) a. [Context: A car dealer stands with a friend in front of a line of cars in his parking lot. He points to them one by one and says:]
 maxarti ___, lo maxarti ___, maxarti ___, maxarti ___, lo maxarti ___...
 sold.1SG not sold.1SG sold.1SG sold.1SG not sold.1SG
 ‘I sold that one, didn’t sell that one, sold that one, sold that one, didn’t sell that one...’
 b. [Context: Earlier today I forwarded a commercial mail to my friend, which I thought she should read. We now meet in a cafe, and without any introduction, I ask:]
 nu, karat ___?
 well read.2SG.F
 ‘Well, did you read it?’

- c. [Context: The earth is rumbling under my feet. I turn in panic to my friend and say:]
 ata margiš ____? ze kmo re'idat adama.
 you feel.SG.M it like shaking earth
 'Do you feel it? It's like an earthquake.'
- d. [Context: In a foreign city, we stop our car and ask for directions. After a lengthy explanation from a local how to get to our destination, my friend reassures me:]
 al tid'ag, katavti ____.
 not worry.FUT.2SG.M wrote.1SG
 'Don't worry, I wrote it down.'

Things are dramatically different if the null object hosts a resumptive pronoun, which functions as the sole variable for an $\bar{A}$ -operator to bind. The BVQ requires this resumptive pronoun to be syntactically present, but deep anaphors provide no internal structure to host it. The result is ungrammatical across a range of $\bar{A}$ -operators – restrictive relative (33a,b), interrogative, (33c) and free relative (33d). More precisely, it is ungrammatical insofar as the intended variable is *internal* to the null object and not the object itself. The examples below are constructed in such a way that binding of the null object produces an anomalous reading (e.g., selling a friend, reading a business, etc.). The fact that such a reading is inescapable supports the conclusion that no alternative analysis is available whereby the variable resides *inside* the null object.

- (33) a. [Context: as in (32a)]
 *yeš li xaver exad še-maxarti ____.
 there.is to.me friend one that-sold.1SG
 ('I have one friend that I sold.')
- b. [Context: as in (32b)]
 *yeš batey-esek še-ani lo koret ____.
 there.is houses.of-business that-I not read.SG.F
 ('There are businesses that I don't read.')
- c. [Context: We're looking at statistical data on the deterioration in students' performance across a range of subjects in high school, and I ask:]
 *eyze mikcoa (lo) margišim ____?
 which subject not feel.PL.M
 ('Which subject does one (not) feel?')
- d. [Context: A cop is writing down the names of witnesses in a crime scene]
 *mi še-ha-šoter katav ____ yuzman la-taxana.
 who that-the-cop wrote.3SG.M will.be.invited.3SG to.the-station
 ('Those people that the cop wrote will be invited to the police station.')

We should now situate these examples in contexts providing a linguistic antecedent for the null object. If, as the AE literature suggests, DPs in languages like Hebrew can undergo ellipsis, they should be elidable in such contexts. DP ellipsis would then give rise to syntactically complex null objects that can host a resumptive pronoun. Consequently, examples (33a-d) would become grammatical. The prediction is fully confirmed, and as expected, the operator in the elliptical clause is indeed interpreted as binding a variable inside the ellipsis site.

- (34) a. *yeš li xaver exad še-kaniti et ha-oto šelo,*
 there.is to.me friend one that-sold.1SG ACC the-car his
 ve-yeš li xaver axer še-maxarti ____.
 and-there.is to.me friend another that-sold.1SG
 ‘I have one friend whose car I bought and I have another one whose car I sold.’
- b. *yeš batey-esek še-ani koret meylim mehem,*
 there.is houses.of-business that-I read.SG.F emails from-them
 ve-yeš batey-esek še-ani lo koret ____.
 and-there.is houses.of-business that-I not read.SG.F
 ‘There are businesses that I read emails from and businesses that I don’t.’
- c. *eyze mikcoa margišim et ha-hidarderut šelo*
 which subject feel.PL.M ACC the-deterioration its
 ve-eyze mikcoa lo margišim ____?
 which subject not feel.PL.M
 ‘Which subject does one feel its deterioration and which subject one doesn’t?’
- d. *mi še-ha-šoter katav et ha-šem šelo*
 who that-the-cop wrote.3SG.M ACC the-name his
 ve-mi še-hu lo katav ____ – zaxu le-yaxas šone me’od.
 and-who that-he not wrote.3SG.M won.3PL to-treatment different very
 ‘Those people whose names the cop wrote down and those he didn’t received very different treatments.’

Differently from (33a-d), the null objects in (34a-d) are surface anaphors (= elided DPs); the operator forms a licit variable binding relation with a resumptive pronoun contained within the elided DP. The syntactic contrast is shown in (35), using the minimal pair (33a) vs. (34a) for illustration (English words used for convenience).

- (35) a. *yeš li xaver exad še-maxarti ____.*
 there.is to.me friend one that-sold.1SG
 (‘I have one friend that I sold.’)

- b. *NCA parse: BVQ violated*
 * There is to me a friend [Op_i that I sold].
 ↳ no variable to bind
- c. *DPE parse: BVQ respected*
 There is to me a friend [Op_i that I sold his_i ear].

 ✓ variable binding

Similarly to the EN-NPE distinction, then, the NCA-DPE distinction systematically distinguishes null positions lacking syntactic structure from those having it. The line of demarcation is exactly the one established in the classic study of Hankamer & Sag 1976 – licensing by pragmatic or linguistic context, while the empirical effects are novel, drawing on the interaction of the BVQ with resumptive pronouns.

4 PP-ellipsis

In section 4.1 I discuss the difficulty in establishing the grammatical nature of missing argumental PPs, given that so often they are just optional, even when I-selected (namely, when the P-head is invariant). Sloppy readings are further found in NCA scenarios, hence they are also unreliable diagnostics for AE of PPs. And extraction from PPs is even more rare than extraction from DPs. Section 4.2 shows that the EIR test can provide the much-needed diagnostic tool. Deep anaphoric null PPs (an instance of NCA) become ungrammatical in environments where an operator occurs with no variable to bind. Once placed in a context that provides a linguistic antecedent to the PP, the ellipsis derivation becomes available. A resumptive pronoun can appear inside the elided PP and satisfy the BVQ, as depicted in (7c).

4.1 The challenge of establishing PP *ellipsis*

The status of PP ellipsis (PPE) is the least clear among the three test cases in this study. Determiners and quantifiers are inherently incomplete without a nominal nucleus, perhaps even uninterpretable. Likewise, many transitive verbs cannot appear without their direct object unless a very specific context (deictic or linguistic) is set up. Thus, it makes perfect sense to analyze “gapped” versions of these constituents (namely, [D/Q ____], [V ____]) as involving some grammatical representation of the missing constituent.

With PPs, however, things are different. They are often optional, even as arguments that are inherently implied by the verb; e.g., observe the contrast *John listened/*heard ____*, which is quite characteristic, even if not absolute. The very property “argumenthood” is not an all-or-none distinction for PPs. While some are strongly argumental (*John depends [PP on Mary]*), others are more properly described as “extended” or “applied” arguments (*John baked a cake [PP for Mary]*). It is certainly not my intention here to delve into these complex questions.

Rather, my modest goal will be to look at the existing arguments for PPE and evaluate their strengths and weaknesses.

As with DPE, a prominent argument for PPE is based on sloppy readings. Saito (2007) cites (36a) from Japanese, and (36b) shows the same in Hebrew.

- (36) a. Taroo-wa zibun-no oya-no ie-ni sundeiru *Japanese*
 Taroo-TOP self -GEN parent-GEN house-in live
 Demo, Hanako-wa sunde-inai ____.
 but Hanako-TOP live-not
 ‘Taroo_i lives in his_i parents' house. But Hanako_j does not live
 in his_i/her_j parents' house.’
- b. A: ravti im aba šeli etmol. *Hebrew*
 quarrelled.1SGwith dad my yesterday
 ‘I quarrelled with my dad yesterday.’
 B: ani bidiyuk hišlamti ____.
 I just reconciled
 ‘I’ve just reconciled with him/mine.’

Null PPs display analogous sloppy readings in Hindi, Bangla and Malayalam (Simpson et al. 2013), in Colloquial Singapore English (Sato 2014), in Korean (Yim 2020) and in Egyptian Arabic (Soltan 2020).

Nonetheless, section 3.1 has demonstrated that sloppy readings as such are not a telltale sign for ellipsis of DPs. The question is whether there is reason to trust them any more for the ellipsis of PPs. In fact, there reason is *not* to.

Just like DP arguments, and even more commonly, PP arguments may undergo NCA. And just like the former supports sloppy readings (see (26)), the latter does too. Note that the verb *naga* ‘touch’ selects a PP in Hebrew, headed by *be-* ‘in, at’.

- (37) [Context: In a spiritual retreat, all participants are expected to touch the one sitting next to them before they start eating. I touch my friend Dan, who is sitting next to me, but I can’t see Mary, who is three seats away. I ask Dan:]
 hi nag’a ____?
 she touched.3SG.F
 ‘Did she touch (the one next to her)?’

More reliable evidence for PPE than sloppy readings exists but is quite sporadic. Cheng (2013:222) and Takahashi (2014:101) cite examples from Chinese and Turkish, respectively, where a null PP argument is understood as a reciprocal bound by the subject. If syntactic reciprocity requires a syntactic representation of the reciprocator, this would indeed favor the

PPE analysis over NCA. Sakamoto (2020:94) argues that possessors inside idiomatic locative PPs in Japanese (e.g., *in Taro's memory*) undergo covert raising out of the PP even when it is null. This is potentially a strong argument for PPE, utilizing the extraction logic; however, it involves intricate assumptions, it is quite narrow in scope and is probably specific to Japanese.

Overall, then, the study of null PPs could greatly benefit from the introduction of a novel test, which can establish PPE beyond reasonable doubt, and is readily extendable to any language with productive resumptive $\bar{A}$ -dependencies.

4.2 Resumptive pronouns inside PPE sites

PP arguments can be omitted quite freely in Hebrew. The verbs in (38a-d) select oblique arguments, headed by the prepositions *al* ‘on’, *le-* ‘to’, *me-* ‘from’ and *be-* ‘in’, respectively. Note that the verbs in (38a,c,d) are different from their English counterparts, which select direct objects. Like the null objects in (32), I assume that the null PPs in (38) are instances of NCA.

- (38) a. [Context: My friend and I approach the city square, where a number of NGOs have lined up their petition booths. My friend tells me:]
 ani lo xotemet ____.
 I not sign.SG.F
 ‘I don’t sign.’
- b. [Context: In a fashion store, I try on a dress, look at myself in the mirror, change back and leave the dress behind. I go to my friend and say:]
 ha-simla ha-zot lo hit’ima ____.
 the-dress the-this not fitted.3SG.F
 ‘This dress didn’t fit.’
- c. [Context: I left some money in my locker and left it unlocked. When I realize this, I rush back with my friend. I open the locker, and my friend, behind me, asks:]
 nu, lakxu ____ ____?
 well took.3PL
 ‘Well, did they take it from you?’
- d. [Context: My friend lent me some trekking gear a while ago. He visits me after my trip and sees the gear exactly where he left it in my house. Nodding at the gear, I say to him:]
 ba-sof lo hištamašti ____.
 in.the-end not used.1SG
 ‘I ended up not using it.’

As we now expect, these null PPs behave quite differently in contexts involving some $\bar{A}$ -operator. By the BVQ, the operator needs a variable to bind, but deep anaphors (like the omitted

PPs) cannot host any internal variable. The result is ungrammatical whether the operator forms a restrictive relative (39a,b), an interrogative (39c) or a free relative (39d). Note that it makes no difference whether a coherent interpretation relies on the resumptive pronoun being the prepositional object itself (39b,c) or some internal (possessive) element inside of it (39a,d).

- (39) a. [Context: as in (38a)]
 * yeš irgunim še-ani lo xotemet ____.
 there.is organizations that-I not sign.SG.F
 ('There are organizations that I don't sign.')
- b. [Context: In a fashion store, we see a customer returning a dress to the display. I want to try it on next. The seller, nodding at the receding customer, whispers to me:]
 * zot iša še-ha-simla ha-zot lo hit'ima ____.
 this woman that-the-dress the-this not fitted.3SG.F
 ('This is a woman that this dress didn't fit.')
- c. [Context: Tomer has been stealing stuff from children's lockers. I try to figure out what exactly happened, so I ask the children:]
 * eyze yeladim Tomer lakax kesef?
 which kids Tomer took.3SG.M money
 ('Which kids did Tomer take money?')
- d. [Context: I needed some trekking gear so a number of my friends brought their gear to my yard. I go over the piles and finally decide:]
 * mi še-ani eštameš ____ ze Yaron.
 who that-I will.use.1SG this Yaron
 ('The one I'll make use is Yaron.')

Finally, to facilitate PP *ellipsis* (a subtype of AE), we test these examples in contexts providing a linguistic antecedent for the null PP. If PPs can *only* go missing by NCA, we expect no different results – binding of a resumptive pronoun inside the null PP should be impossible. On the other hand, if PP ellipsis is a genuine option in the grammar, as previous studies have occasionally suggested but failed to establish conclusively, then the result would be grammatical. The latter prediction is indeed confirmed; the operator in the elliptical clause is interpreted as binding a variable inside the ellipsis site (antecedent PPs are in bold). Such cases were first documented in Vardi 2022 (though the data below are my own).

- While the null PPs in (39a-d) are deep anaphors, those in (40a-d) are surface anaphors. Thus, the operator can form a licit variable binding relation with a resumptive pronoun inside the latter but not inside the former. The syntactic contrast is shown in (41), using the minimal pair (39a) vs. (40a) for illustration (English words used for convenience).

- [illegible]

c. *PPE parse: BVQ respected*

There are organizations [Op_i that I don't sign ~~on their_i~~ petitions].

✓ variable binding

Our analysis makes one further, interesting prediction. All the resumptive pronouns discussed above occur inside PPs (or KPs, if *šel* 'of' is K). If PPs are eligible to ellipsis in Hebrew, those containing just the resumptive pronoun should be no exception. The result of eliding a resumptive PP would *appear* to violate the BVQ insofar as the operator would have no visible variable to bind. But this would be an illusion: the elided PP is syntactically present and so is the internal resumptive pronoun. In contrast, this configuration *should* be banned by the BVQ if the PP is dropped by NCA.

With this much in mind, consider the following pair, modified from (20b)/(21b) only in retaining an overt nominal head (*sfarim* 'books') after the partitive Q.

(42) a. [Context: Looking at a pile of Continental Philosophy books on my friend's desk, I sigh and say:]

*yeš mexabrim še-maspik li likro rak xelek me-ha-sfarim.
there.exist authors that-enough to.me to.read only part from-the-books
(‘There are authors that it’s enough for me to read only part of these.’)

b. yeš mexabrim še-xašuv li likro et kol **ha-sfarim šela’hem**,
there.exist authors that-important to.me to.read ACC all the-books their
ve-yeš mexabrim še-maspik li likro rak xelek me-ha-sfarim.
and-there.exist authors that-enough to.me to.read only part from-the-books
‘There are authors that it’s important for me to read all their books,
and there are authors that it’s enough for me to read only part (some) of the books.’

Focusing on the relevant elements and using English words for convenience, the structure of (42b) is as follows.

(43) Antecedent clause: ... *authors* [Op_i [... *all the books of.them_i*...]]

Elliptical clause: ... *authors* [Op_i [... *part from-the-books of.them_i*...]]

The possessive PP/KP in the first clause licenses (optional) ellipsis of its counterpart in the second clause. The only interpretation available is for the QP *part of the books* to be understood as being written by the authors mentioned in the second clause. This obligatory interpretation is mediated by the silent resumptive pronoun, which must provide the variable for the operator associated with that second occurrence of *authors*.

Once again, we see that the test of ellipsis-internal resumption reliably distinguishes deep from surface anaphora, this time within the class of null PPs. This result is especially valuable in

light of the considerable unclarity in the literature as to whether “optional PPs” can truly emerge from bona fide ellipsis.¹¹

5 Implications for “mixed anaphors”: Beyond the extraction test

Sections 2-4 have established a major contribution of the EIR test: It can demonstrate the existence of surface anaphora at a level of confidence comparable to that of the extraction test, even when applied to domains opaque to extraction. In this section I turn to a second contribution: The EIR test can refine our understanding of what exactly goes wrong in cases where extraction fails out of a null domain which, in principle, is *not* opaque to extraction. This is the characteristic pattern of the so-called “mixed anaphors”, mentioned in section 1. Before turning to them, let me highlight one broader implication that touches on a lingering issue in discussions of deep and surface anaphors.

This lingering issue concerns the proper treatment of NCA. It has already been hinted in Hankamer & Sag 1976:fn. 25, and then was fully expressed in van Craenenbroeck and Merchant 2013: 710 and Depiante 2019:678: Might NCA be syntactically complex after all, except we cannot see this?¹² Following Aelbrecht’s (2010) timing-based account of why $\bar{A}$ -extraction is impossible out of a Dutch *surface* anaphor (Modal Complement Anaphora), van Craenenbroeck and Merchant observe: “It could well be that even null complement anaphora involves abstract syntax, but that the timing of the ellipsis process in this specific construction precludes any movement operation from targeting material inside the ellipsis site.” Depiante proceeds to speculate that “all null anaphors are the product of deletion. Then, the difference between deep and surface anaphors will be lost as a theoretical primitive and would be reduced to the timing of the deletion operation.”

Because the EIR test is *not* timing-sensitive, and only responds to the well-formedness of LF-representations, it is a better probe into this question than plain extraction. No matter how early the putative deletion of NCA would occur, its syntactic structure and semantic content should be fully visible at LF. An internal resumptive pronoun should then be available for any operator

¹¹ In (39)-(40), the predicate selecting the null PP is a verb. Vardi (2022) found that the contrast gets blurred when the predicate is an adjective (e.g., *acbanī al X*, ‘angry on X’, *kašur le-X* ‘tied to X’). Specifically, adjectival passives yield worse results than verbs in environments like (40a-b), and simple adjectives are even worse, although there is considerable inter-speaker variation. Vardi hypothesized that the argumental status of PP complements of adjectives is less rigidly grammaticalized, so that (for some speakers) those PPs are treated as adjuncts, an option more common with adjectives unrelated to verbs. Because PPE is a species of AE, and AE does not apply to adjuncts, adjunct null PPs cannot arise by PPE; hence, they must involve NCA. Alternatively, adjectives may be ambiguous with regard to taking arguments. Note that even adjectives that appear to select a PP argument need not imply it when it is absent, leaning towards an individual-level reading (e.g., *John was proud/angry* does not entail that there is an X s.t. John was proud of X or angry at X). The absence of a PP argument, at least in some circumstances, explains the resistance to resumptive dependencies, which depend on a full syntactic structure in the null position.

¹² As distinct from the question of whether NCA is syntactically present. It is possible for NCA to be present without being complex (e.g., as a *pro*); see Ranero and Royer 2024 for a recent discussion.

in need of such. Clearly, this is *not* what we find. The Hebrew data in sections 2-4, and data from English, Dutch, Danish and Korean discussed below, indicate that there is a genuine class of anaphors containing no internal structure at LF at all, hence they systematically fail the BVQ. I conclude from that finding that the deep-surface anaphor distinction is not reducible to the derivational timing of deletion.

These considerations underscore the inherent limitations of the extraction test. Let us run this argument more explicitly. Consider an abstract scenario wherein an operator can be extracted from a constituent XP when XP is overt but not when it is null.

(44) *Failed extraction out of null transparent domains*

- a. ✓ No ellipsis: $Op_i \dots [XP \dots t_i \dots]$
 b. * Ellipsis: $Op_i \dots [XP \dots ? \dots]$
- $\swarrow \quad \searrow$
 c. * $Op_i \dots [XP \dots t_i \dots]$ d. * $Op_i \dots [e]$

There are, in fact, two logically possible reasons for the “*” in (44b). Either some property of ellipsis interacts with some property of movement during the derivation, with the lethal outcome that one of them bleeds the other, as in (44c).¹³ Or else, the null XP may turn out to be a deep anaphor (*pro* or not even syntactically present), as in (44d), in which case the structure is ruled out by the BVQ.

Although the extraction test is often put to use in the ellipsis literature, it is rarely recognized that it is inherently confounded (see Depiante 2019 for an exception). While *success* of extraction unambiguously points to a surface anaphor, *failure* of extraction is open to either interpretation (44c) or (44d). Unless the nature of the null category is presupposed in advance, failure of extraction simply cannot decide the fundamental issue of deep vs. surface anaphora.

It is here that the EIR test can claim a distinctive advantage over the extraction test. Suppose that extraction fails out of a null XP, and we attempt internal resumption, and it fails too. Then we can safely conclude that (44d) is warranted: The null XP is genuinely a deep anaphor, and as such does not care how the internal variable comes into being (via movement or resumption), because there is just no variable. Suppose the EIR test yields a grammatical sentence, while extraction is still out. Then we can safely endorse (44c): Some derivational clash specifically

¹³ Commonly in the literature on derivational ellipsis it is ellipsis that bleeds $\bar{A}$ -extraction (a.o., Baltin 2012, Aelbrecht 2010); see Landau 2020 for the opposite case, where it is movement (specifically, head movement) that is claimed to bleed ellipsis.

rules out *extraction* combined with ellipsis, but has no bearing on resumption combined with ellipsis, which is licensed at LF. The null XP is thus a surface anaphor.

Either conclusion may be true of different constructions. I now illustrate the first line of argumentation with a number of test cases: The mixed anaphors English *do so*, Dutch *dat doen* ‘do that’ and Danish *det*, and Korean null objects, whose analysis is currently contested.

Consider first the notorious troublemaker, English *do so*. An extensive literature on this single construction is divided between the deep anaphor analysis (Kehler and Ward 1999, Houser 2010, Breuning 2019) and the surface anaphor analysis (Hankamer & Sag 1976, Baltin 2012, Hellman 2013, Kim 2016, Park 2018). My modest goal here is not to deliver an ultimate verdict on this venerable dispute, but merely to adduce one novel piece of evidence *against* the surface anaphor analysis.

As is well-known, $\bar{A}$ -extraction is impossible out of *do so* anaphora, in sharp contrast to VP-ellipsis (Breuning 2019:2).

- (45) a. I know which cases the Supreme Court will hear, and which cases they won’t.
b. Bubble tea, I’ll drink, but green tea, I won’t.
c. * I know which cases the Supreme Court will hear, and which cases they won’t do so.
d. * Bubble tea, I’ll drink, but green tea, I won’t do so.

On the analysis of *do so* as a surface anaphor, something special must be said here. The natural and popular response is that something about how $\bar{A}$ -movement proceeds is incompatible with ellipsis. The problem may be one of bleeding, $\bar{A}$ -movement being triggered “too late”, after ellipsis had removed the VP (or rendered it inaccessible); or conversely, if ellipsis is LF-copying, a counter-feeding problem: $\bar{A}$ -movement being triggered too early, before the VP-structure is filled; or a locality problem involving phase impenetrability (see Baltin 2012, Hellman 2013, Kim 2016 and Park 2018 for different implementations of these options).

Crucially, all these proposals attribute the failure of extraction to some *derivational* property of $\bar{A}$ -movement, that is, to a property emerging prior to LF. It is therefore predicted that $\bar{A}$ -dependencies that are established only at LF, with no intermediate derivational steps, should not lead to the lethal interaction with ellipsis. Resumptive dependencies are exactly that: They involve no movement, only coindexation, which is interpreted as λ -binding in the semantics, and may not even be formally marked prior to LF. English only uses resumptive pronouns comfortably in Left Dislocation, so we may test whether *do so* anaphora can host the resumptive pronoun that is bound by the left-dislocated element.

What we find, in fact, replicates the finding from $\bar{A}$ -extraction: The dependency can cross VP-ellipsis but not *do so* anaphora.

- (46) a. Deep Purple, I never bought their records. Led Zeppelin, I did.
 b. My former landlord, I didn't keep his apartment clean. The current one, I do.
 c. * Deep Purple, I never bought their records. Led Zeppelin, I did so.
 d. * My former landlord, I didn't keep his apartment clean. The current one, I do so.

On the surface anaphor analyses of *do so*, (45c-d) and (46c-d) would receive independent explanations; the latter cannot be explained by reference to properties of movement. On the other hand, if *do so* is a deep anaphor, all these cases receive a unified explanation: they all violate the BVQ. This constraint does not distinguish between operators associated with movement and those associated with base-generated dependencies (see (3)-(4)). Indeed, it only scans LF representations and nothing that precedes them, which seems like the proper level of generalization for the facts at hand.¹⁴

A second test case, quite similar in nature, is the Dutch mixed anaphor *dat doen* 'do that'. According to Henyen 2022, *dat doen* blocks most types of extraction: Passive A-movement, QR and standard $\bar{A}$ -movement. It does allow unaccusative A-movement and comparative $\bar{A}$ -movement (of a degree operator). Henyen ultimately analyses it as a surface anaphor; the impossible extractions are accounted for by independent derivational mechanisms of locality, although the passive-unaccusative contrast is left unexplained.

When tested for EIR, *dat doen* yields ungrammatical results (Marcel den Dikken, p.c.).

- (47) a. * Jan, ik heb zijn muffins al eens geproefd;
 Jan I have his muffins already once tasted
 maar Wim, ik heb dat nog niet gedaan.
 but Wim I have that not yet done
 Lit. 'John, I have already tasted his muffins once, but Wim, I haven't done that.'
 b. * Ann, ik kon de brieven van haar wel vinden;
 Ann I could the letters from her AFF find
 maar Mary, ik heb dat niet kunnen doen.
 but Mary I have that not can.INF do
 Lit. 'Ann, I could find the letters from her, but Mary, I couldn't do that.'

I take these data to be supportive of solution (44d): *dat doen* reveals no internal structure at LF, hence it is a deep anaphor. The grammatical unaccusatives perhaps should receive a movement-

¹⁴ Unlike $\bar{A}$ -dependencies, A-dependencies do cross *do so* anaphora; see Breuning 2019 for an analysis of these facts with no recourse to A-movement.

free analysis, along the lines of Breuning 2019, while the comparative deletion data remain an open problem.

The third test case is the Danish mixed anaphor *det*, which allows A-extractions but not $\bar{A}$ -extractions (Houser et al. 2007). On Houser et al.’s account, *det* is a surface anaphor; $\bar{A}$ -extraction of VP-internal material is blocked due to locality, specifically of the A-over-A type, since the VP itself, spelled out as *det*, is a candidate for $\bar{A}$ -movement to [Spec,CP]. Notice that this derivational account does not carry over to resumptive dependencies, which are not derived by movement. Hence, it predicts anaphoric *det* to pass the EIR test. The prediction, however, is disconfirmed (Line Mikkelsen, p.c.).

- (48) a. Den gamle bager_i, jeg har smagt *(hans_i) rugbrød.
 the old baker I have tasted his rye.bread
 ‘The old baker_i, I have tasted his_i rye bread.’
 b. * Den gamle bager, jeg har smagt hans rugbrød,
 the old baker I have tasted his rye.bread
 men den nye bager, jeg har det ikke.
 but the new baker, I have it not
 (Intended: ‘The old baker_i, I have tasted his_i rye bread, but the new baker,
 I haven’t.’)

As expected, the BVQ mandates a resumptive pronoun in standard Left Dislocation (48a). A parallel resumptive pronoun, however, is not recoverable from the VP-anaphor *det* in (48b). Similarly to English *do so* and Dutch *dat doen*, then, Danish *det* displays the characteristic behavior of a deep anaphor, associated with no internal structure at LF.

Our fourth and final test case concerns null objects in Korean. There is an ongoing debate among Korean syntacticians whether all instances of such null objects are explicable in terms of *pro* or whether AE (DP ellipsis) is also required.¹⁵ The crucial extraction test is under debate: While some authors argue that Korean null CP arguments block extraction (Saito and An 2010, Ahn & Cho 2011b, 2022), others claim they allow it, either covertly (Sakamoto 2020) or overtly (Park to appear). Without entering the complexities of testing extraction, we can ask whether the EIR test can shed light on the analysis of Korean null arguments.

As shown in (49a), in an out-of-the-blue context, Left Dislocation in Korean requires the presence of an overt resumptive pronoun. This baseline provides us with evidence that the BVQ

¹⁵ For the *pro* analysis, see Park 1997, Ahn and Cho 2010, 2011b, 2011a, 2020, 2021, 2022, 2024, Kim 2010, Moon 2010, 2019, Bae and Kim 2012 (see Park and Lee 2025 for a bare N analysis, similarly lacking internal structure). For the AE analysis, see Kim 1999, Saito 2007, Saito and An 2010, Lee and Kim 2010, Um 2011, Takahashi 2014, Lee 2016, Sakamoto 2020, Han et al. 2020, Yim 2020, Park to appear.

is operative in such constructions. Interestingly, when the RP-containing object remains null, the sentence is ruled out (Heejeong Ko. p.c.).

- (49) a. Na-uy i chinkwu, nay-ka *(ku-uy) cha-lul sasse.
 I-GEN this friend I-NOM he-GEN car-ACC bought
 ‘This friend of mine, I bought *(his) car.’
- b. *Na-uy i chinkwu, nay-ka ku-uy cha-lul sasse.
 I-GEN this friend, I-NOM he-GEN car-ACC bought
 Nay-uy ce chinkwu, nay-ka phalasse ____.
 I-GEN that friend I-NOM sold
 (Intended: ‘This friend of mine, I bought his car. THAT friend of mine, I sold his car.’)

The ungrammaticality of (49b) is unexpected if the null object is a surface anaphor with internal structure hosting a resumptive pronoun. Thus, this finding seems to lend support to the line of research analysing Korean null objects as *pro* rather than the outcome of AE. Evidently, further research should address these facts and integrate them with the larger body of data involving Korean null objects.

Recall that the present analysis predicts the possible occurrence of cases like (44c): A given ellipsis site blocks subextraction due to some derivational clash, but does allow internal resumption. Unfortunately, so far I have been unable to identify data of this type. Possibly, this is due to the combined rarity of timing-sensitive, extraction-blocking ellipsis and productive resumption strategies across languages. However, the existence of such cases is a logical implication of the present analysis, and further research would hopefully unearth them.

6 Conclusion

Evidence for silent syntactic structure can be broadly divided into interpretive or syntactic evidence. Interpretive evidence consists in readings assigned to the null position which presumably go beyond the expressive scope of simplex, unstructured anaphoric elements (NCA, *pro*, $\bar{A}$ -variable). Decades of intensive research, however, have shown that such readings are frustratingly sparse and too frail to support the fundamental, architectural distinction between deep and surface anaphora. Syntactic evidence, in turn, mainly consists in agreement or movement dependencies crossing the ellipsis boundary. The problem is that quite often it is unavailable due to independent factors that either block the formation of the dependency or make its outcome invisible.

The main goal of this paper has been to develop and put to use a novel syntactic test, the EIR, whose outcome can decide the issue of deep vs. surface anaphora as conclusively as the agreement/extraction tests do. The EIR test solely rests on the BVQ, a syntactic condition on the well-formedness of LF representations; it is indifferent to derivational interactions or islandhood effects which severely restrict the applicability of agreement into and extraction out of ellipsis sites.

Assisted by that test, we have been able to demonstrate how productive elliptical processes are in Hebrew. Although the language employs various strategies of deep NP-, DP- and PP-anaphora, it also allows each of these categories to undergo ellipsis in appropriate contexts; precisely there but not in the deep anaphora contexts, resumptive pronouns are licensed inside the anaphoric site and can be bound by operators outside of it. The discussion has highlighted a number of pitfalls in establishing the reality of ellipsis in these domains, thus hopefully refining the methodological toolbox of ellipsis scholars.

Finally, the results of this study can be brought to bear on some recalcitrant cases of “mixed anaphors” by informing the debate on their nature, possibly deciding whether they are simple deep anaphors or unusually constrained surface anaphors. The EIR test is especially useful in this regard thanks to its relative independence of “interfering” factors; it provides a nearly pure window to LF. A number of such cases were discussed in section 5, but clearly we have only touched the tip of the iceberg there. A novel test calls for novel testing grounds, and these will surely emerge in the future.

Appendix: Resumption in Hebrew is not derived by movement

Early studies of resumption in Hebrew assumed that resumptive pronouns (RPs) are bound by base-generated operators, that is, the dependency was not derived by movement (Doron 1982, Borer 1984, Sells 1987, Erteschik-Shir 1992, Shlonsky 1992, Arad 2014). An alternative view, however, has developed more recently, by which obligatory RPs in Hebrew – specifically oblique and genitive ones, as in (6), in non-island environments – are (PF-generated) residues of genuine syntactic movement (Sichel 2014, 2021, 2022, Rasin 2017, Bassi & Rasin 2018). Importantly, on anyone’s view, *some* RPs in Hebrew are never generated by movement – namely, optional RPs (i.e., in direct object and non-local subject positions) and RPs inside islands. At a minimum, then, examples like (21c), where the RP is located inside a subject island, and (40a,c,d), where the RP is located inside the complement of a PP island, unequivocally show that base-generated RPs furnish reliable evidence for surface anaphora in environments not amenable to extraction. The debate is confined to the status of obligatory RPs in non-island environments.

I claim that there is no convincing empirical basis for distinguishing between the two classes of RPs. More specifically, I will show that purported movement-related properties attested with

obligatory RPs in non-island environments are also attested, at least sometimes, with optional RPs and with RPs inside islands.

- (50) a. Class I: Obligatory RPs in non-island environments.
Class II: Optional RPs and RPs inside islands.
b. Claim: Class I RPs and class II RPs are syntactically nondistinct – both related to their $\bar{A}$ -binders by binding, not by movement.

The case for distinguishing between the two classes entirely rests on alleged differences in reconstruction, which in turn is taken to be a hallmark of movement dependencies.

- (51) *The argument in Sichel 2014, 2011, 2022, Rasin 2017 and Bassi & Rasin 2018*
a. Empirical claim: Class I RPs allow reconstruction, class II RPs block it.
b. Theoretical claim: Reconstruction is only possible in movement dependencies.
c. Analytic conclusion: Class I RPs residues of movement, class II RPs are not.

My challenge to this argument will be twofold. First, I will show that the empirical claim (51a) is false: Class II RPs do allow reconstruction of various types. Indeed, Sichel (2014:668, fn. 11) acknowledges that “for some speakers, judgments were not always consistent”. It is a classic finding that preferences for a gap or for an RP are affected by a variety of processing factors, many of which fall outside core grammar (Erteschik-Shir 1992, Ariel 1999, McKee and McDaniel 2001, Alexopoulou and Keller 2007). I believe that such preferences have confounded the reconstruction data, falsely creating the illusion that there is a class of RPs which consistently and categorically blocks reconstruction.

Second, I will argue that the theoretical claim (51b) cannot be correct either, since incontrovertible non-movement dependencies (resumptive and others) also exhibit reconstruction. This point is actually not contested by the above authors. While they all agree that there is more than one path to reconstruction, they simultaneously rely on (51b) in drawing the conclusion in (51c) (for example, this quote from Sichel 2022: 9: “The availability of reconstructed readings suggest that the derivation must involve head raising”). If one drops the assumption that reconstruction data teach us something about *movement*, (51c) remains unsupported.¹⁶

I claim that (51c) must be rejected in favor of the unitary analysis in (50b). The outline of my argument thus follows, nearly point by point, the argument made extensively in Hewett 2023 for Arabic dialects (Iraqi, Syrian and Tunisian), and for Jordanian Arabic and Mandarin

¹⁶ Interestingly, Sichel (2014:677) does not rule out the theoretical option of RPs (including optional ones) inhabiting a Matching-analysis of relative clauses, in effect allowing reconstruction. Rather, she states that the question is not decidable. But in fact, it is. Optional pronouns that allow reconstruction *require* a Matching analysis even on Sichel’s assumptions. If they further allow reconstruction inside islands, as shown below, Matching must be available independently of movement. This has the consequence of making a movement derivation of RPs completely redundant, hence dispensable.

Chinese elsewhere (see Hewett 2023:319-320). In work in progress, I go into further detail and theoretical issues surrounding the analysis of resumption. The narrow goal of this appendix is simply to show that reconstruction data cannot establish that RPs in Hebrew, of any type, are derived by movement (while non-movement-derived RPs, to recall, are indispensable).

Notably, this conclusion has already been thoroughly defended by Arad (2014), who has shown that Hebrew RPs fail to license parasitic gaps *even* when they support reconstruction. Arad further showed that reconstruction is available even in scenarios forcing an external RC-head. Her conclusion was that Hebrew RPs (and, in fact, traces too) avail themselves of *semantic* reconstruction. While the Matching analysis sketched below in (57) makes semantic reconstruction unnecessary, I fully share Arad’s conviction that reconstruction data do not decide the movement-nonmovement question in the analysis of RPs.

The examples below illustrate reconstruction for reflexive binding (where the reflexive is contained in the relative head), idiom chunks, amount relatives and free relatives; the latter two types are commonly assumed to *require* reconstruction for their proper interpretation (Grosu and Landman 1998). Data involving *de dicto* readings and inverse scope are discussed elsewhere, in work in progress, but are omitted here as they involve certain complications that would take us too far afield (see Sharvit 1999 and Guilliot and Malkawi 2007, respectively, for pertinent discussion).

Reflexive binding

Sichel’s (2014) examples that test reflexive binding under reconstruction all use the NP *šmu’a al acmo* ‘rumor about himself’. However, NP-internal reflexives are quite uncommon in Hebrew in comparison to English. In fact, this specific NP, *šmu’a al acmo*, does not appear in Google search even once. More common is *tmuna šel acmo* ‘picture of himself’. When heading a relative clause that contains a binder for the reflexive, this NP can be reconstructed to its base position, whether it hosts a gap or an accusative RP (here and below, to ensure naturalness, all examples with RPs place them remotely from the relative head; boldfacing is intended to highlight the RP, not to mark any phonological stress).

- (52) *atem zoxrim et ha-tmuna šel acma_i še-kulam hayu betuxim*
 you.PL remember.PL.M ACC the-picture of herself that-all were sure.PL.M
še-Kim Kardashian_i ziyfa (ota)?
 that-Kim Kardashian faked.3SG.F it
 ‘Do you all remember the picture of herself that everybody was sure that
 Kim Kardashian faked?’

Idiom chunks

Sichel (2014:659, fn. 1) recognizes that the relevant idiom data are not decisive. In fact, it is not difficult to construct examples with optional RPs resuming an idiom chunk insofar as that idiom chunk can be relativized to begin with (as is well-known, idioms are not all alike, and the more decomposable ones lend themselves more easily to syntactic separation; see Fadlon et al. 2018). In the example below, the RP occurs at an embedded subject position, which belongs to class II (namely, an optional RP). The idiomatic reading is accessible either with a gap or with the RP.

- (53) sof sof hitxil diyun al ha-kvisa ha-meluxlexet še-mi-kol
end end began.3SG.F discussion on the-laundry the-dirty that-from-all
ha-pirsumim kvar hevanu še-(hi) yac'a ha-xuca be-xul.
the-publications already understood.1PL that-it went.out.3SG.F the-outside in-abroad
'Finally a discussion began about the dirty laundry that was washed abroad,
as we've already understood from all the publications.'

Amount relatives

Grammatical examples of amount relatives with accusative RPs can be found on the internet, as the following pair illustrates (the gap version is of course grammatical too).¹⁷

- (54) a. merxak še-af exad pašut lo yodea eyx la'avor (oto).
distance that-no one simply not know.SG.M how to.pass it
'A distance that simply no one know how to pass.'
b. ze yaxol lefanot zman yakar še-nitan
this can.SG.M to.free.up time precious that-possible
levalot (oto) im ha-mišpaxa.
to.spend it with the-family
'This can free up precious time to spend with the family.'

Note that the RP in (54a) is located inside an infinitival *wh*-complement – a (very) weak island in Hebrew. I return shortly to resumption inside islands.

Free relatives

Borer (1984:239) claimed that free relatives (FRs) in Hebrew reject accusative RPs. Her single example, however, tested a FR headed by *ma* 'what'. Sichel (2014:662) follows that claim, although she only assigned a marginal status ("??") to her FR example with an accusative RP. Significantly, Sichel tested FRs headed by *mi* 'who'. In fact, there is a contrast between the two

¹⁷ https://simania.co.il/showReview.php?reviewId=123961#google_vignette
<https://invitro.org.il/career-and-family-life/>

types: while *mi* can be resumed in FRs for all speakers, only a subset of speakers allow resumption of *ma*. Natural *mi*-examples are easy to find on the internet, illustrating RPs inside FRs, in positions that alternate with a gap, hence predicted to reject RPs on Sichel's account.¹⁸

- (55) a. lexu lenaxem et mi še-ha-medina natša (oto).
 go.IMP.PL to.console ACC who that-the-state abandoned.3SG.F him
 'Go console those abandoned by the state.'
- b. tivxeru be-mi še-atem xošvim še-(hu) mat'im.
 will.choose.2PL in-who that-you.PL think.PL.M that-he suitable.SG.M
 'Choose whoever you think is suitable.'

While the sensitivity of resumption in FRs to the [$\pm$ human] distinction calls for explanation (one such explanation is offered in work in progress), (55) clearly shows that optional RPs *can* be reconstruction sites in Hebrew FRs.

Reconstruction inside islands

Previous work has established that, in quite a few languages, resumption is compatible with reconstruction even inside islands (Guilliot and Malkawi 2007, 2009, 2011, Malkawi and Guilliot 2007, Salzmann 2017; see the extensive literature review in Hewett 2023:313). The relevant examples are inevitably complex and unnatural, but reconstruction as such does not seem to make them any worse. Hebrew is no different. (56a) is constructed with the idiom *hipil tik al X*, literally "drop a file/bag on X", which means 'impose an annoying/difficult task on X'. The idiom chunk *tik* 'file/bag' is the relative head, binding an accusative RP, which occurs inside a Complex NP island. (56b-c) are examples from the internet.¹⁹ In the FR of (56b), the relative pronoun *mi* 'who' binds a nominative RP occurring inside a temporal adjunct. In the amount relative of (56c), the relative head *calories* is understood as *amount of calories*, and it binds an RP inside a conditional adjunct. All these adjuncts are islands for extraction.

- (56) a. ze tik še-hayiti muda la-sikun še-yapilu oto alay.
 this file/bag that-was.1SG ACC aware to.the-risk that-drop.FUT.3PL it on.me
 'This is an annoying task that I was aware of the risk of it being imposed on me.'

¹⁸ <https://www.facebook.com/watch/?v=157567386307062>
<https://www.ynet.co.il/articles/0,7340,L-2302493,00.html>

¹⁹ <https://www.myforum.co.il/showthread.php/20149-The-revenge-of-the-princess-from-HELL!-%D7%9E%D7%A7%D7%95%D7%9D-1>
<https://www.chef-bari.co.il/%D7%A7%D7%9C%D7%95%D7%A8%D7%99%D7%95%D7%AA-%D7%9B%D7%9C-%D7%9E%D7%94-%D7%A9%D7%A8%D7%A6%D7%99%D7%AA%D7%9D-%D7%9C%D7%93%D7%A2%D7%AA/>

- b. mi še-axrey še-**hu** af mamšix le'hafria
 who that-after that-he flies.3SG.M continues.SG.M to.interrupt
 la-seder šel ha-misxak.
 to.the-order of the-game
 'A person who, after he's kicked out, continues to disrupt the order of the game.'
- c. gam oxel bari yoter ve-gam oxel bari paxot mexilim
 also food healthy more and-also food healthy less contain.PL.M
 kaloriyot še-im noxal yoter miday me-**hen** ze yigrom
 calories that-if will.eat.1PL too much from-them this will.cause.3SG
 le-xax še-na'ale ba-miškal.
 to-such that-will.rise.1PL in.the-weight
 'Both healthy and unhealthy food contain calories that if we eat too much of them
 it will make us gain weight.'

In sum, contrary to some earlier descriptions, optional RPs in Hebrew allow reconstruction just like obligatory ones do. In this respect, Hebrew patterns with Arabic dialects (Iraqi, Tunisian, Syrian, Jordanian, Lebanese), as well as with Mandarin Chinese (see Hewett 2023:319-320), all of which associate optional and obligatory RPs with the same reconstruction profile. Furthermore, Hebrew RPs allow reconstruction inside islands, again an attested possibility in other languages. Taken together, these facts leave the claim that some RPs in Hebrew are residues of movement with no independent support. Instead, a simple, “traditional” account is all one seems to need – the unitary analysis in (50b).

Obvious questions arise. Chief among them is: How does reconstruction come about without movement? A plausible account, developed in Guilliot and Malkawi 2007, 2009, 2011, Malkawi and Guilliot 2007, Salzmann 2017 and Hewett 2023, invokes the Matching analysis of RCs combined with E-type style NP-ellipsis, as schematized below (English words used for convenience).

(57) *Resumption: Matching + NP-ellipsis*

- a. The movie that I watched it.
- b. [DP the [NP movie] [CP [DP D_[Op,E] [NP ~~movie~~]]_i that I watched [DP D_[Pron,E] [NP ~~movie~~]]_i]
-

On this analysis, the external NP is matched by two identical NPs inside the RC: One in [Spec,CP], associated with the operator D, and one in the base position, associated with the pronominal D (=RP). Both D heads carry the ellipsis-inducing feature [E], which guarantees identity among the three NPs. The presence of the lowest NP copy ensures reconstruction in

much the same way that the elided NP content of E-type pronouns accounts for their interpretation (Elbourne 2001, 2008), as discussed in section 3.1.

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